UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 30/01/2024*

PROBLEM

We observe N samples (for n=0, 1,…, N-1, with N>>1) of a realization of the random process $X[n] = s[n] + W[n]$, where $s[n] = A + B\cos(2\pi F_0 n)$ is the signal of interest (SOI), A and B are the amplitudes of the SOI modelled as unknown deterministic parameters, and $W[n]$ is a zero-mean white Gaussian process with power σ_W^2.

(1) Derive the signal to noise ratio (SNR) $\gamma \triangleq E\left\{\|\mathbf{s}\|^2\right\} / E\left\{\|\mathbf{W}\|^2\right\}$, where **s** and **W** are the vectors containing the N samples of signal and noise, respectively. Then, derive the log-likelihood function, $LLF(A, B, \sigma_W^2) = \ln f_X(\mathbf{x}; A, B, \sigma_W^2)$, where $f_X(\mathbf{x}; A, B, \sigma_W^2)$ is the probability density function (PDF) of the observed data vector $\mathbf{X} = \begin{bmatrix} X[0] & X[1] & \cdots & X[N-1] \end{bmatrix}^T$.

(2) Derive the joint Maximum Likelihood (ML) estimators of the signal parameters A and B when the noise power σ_W^2 is known and provide a geometrical interpretation of the estimators.

(3) Derive the bias and the mean square error (MSE) of the ML estimators of A and B, establish if they are unbiased and consistent, derive which one of the two has better estimation accuracy and explain the reason. Then, plot the MSE of $\hat{B}_{ML}$ as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-3} when the noise power is σ_W^2 =1.

(4) Find the Cramér-Rao bounds (CRBs) on the joint estimation of A and B and establish if the ML estimators previously derived are efficient or not.

(5) Assume now that also the noise power σ_W^2 is unknown: derive an unbiased estimator of σ_W^2.

Solution.

(1) *Derive the SNR and the log-LF.*

$$s[n]=A+B\cos\left(2\pi F_0 n\right) \quad \Rightarrow \quad \mathbf{s}=A\mathbf{h}_1+B\mathbf{h}_2=\left[\mathbf{h}_1 \quad \mathbf{h}_2\right]\begin{bmatrix}A\\B\end{bmatrix}=\mathbf{H\theta}$$

$$E\left\{\|\mathbf{s}\|^2\right\}=\|\mathbf{s}\|^2=\|A\mathbf{h}_1+B\mathbf{h}_2\|^2=A^2\|\mathbf{h}_1\|^2+B^2\|\mathbf{h}_2\|^2, \quad \text{since } \mathbf{h}_1^T\mathbf{h}_2=0, \ \text{ i.e. } \mathbf{h}_1\perp\mathbf{h}_2,$$

In fact,

$$\mathbf{h}_1^T\mathbf{h}_2=\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n\right)\cong 0 \quad (if \ N\gg 1)$$

$$E\left\{\|\mathbf{s}\|^2\right\}=A^2\|\mathbf{h}_1\|^2+B^2\|\mathbf{h}_2\|^2=A^2N+\frac{B^2N}{2}=\left(A^2+\frac{B^2}{2}\right)N, \ E\left\{\|\mathbf{W}\|^2\right\}=N\sigma_W^2,$$

$$\gamma\triangleq E\left\{\|\mathbf{s}\|^2\right\}\Big/E\left\{\|\mathbf{W}\|^2\right\}=\frac{1}{\sigma_W^2}\left(A^2+\frac{B^2}{2}\right)$$

$$X[n]\in N\left(s[n],\sigma_W^2\right), \quad \left\{X[n]\right\}_{n=0}^{N-1} \text{ indipendent} \quad \Rightarrow \quad \mathbf{X}\in N\left(\mathbf{H\theta},\sigma_W^2\mathbf{I}\right)$$

PDF in scalar format:

$$\begin{aligned} f_X(\mathbf{x};A,B,\sigma_W^2)&=\prod_{n=0}^{N-1}f_X(x[n];A,B,\sigma_W^2)\\ &=\left(\frac{1}{\sqrt{2\pi\sigma_W^2}}\right)^N\exp\left(-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(x[n]-A-B\cos\left(2\pi F_0 n\right)\right)^2\right)\end{aligned}$$

PDF in vector format:

$$\begin{aligned} f_X(\mathbf{x};A,B,\sigma_W^2)&=\frac{1}{\left(2\pi\right)^{N/2}\sqrt{\left|\sigma_W^2\mathbf{I}\right|}}\exp\left(-\frac{1}{2}\left(\mathbf{x}-\mathbf{H\theta}\right)^T\left(\sigma_W^2\mathbf{I}\right)^{-1}\left(\mathbf{x}-\mathbf{H\theta}\right)\right)\\ &=\frac{1}{\left(2\pi\right)^{N/2}\left(\sigma_W^2\right)^{N/2}}\exp\left(-\frac{1}{2\sigma_W^2}\|\mathbf{x}-\mathbf{H\theta}\|^2\right)\end{aligned}$$

The log-likelihood function is given by:

$$LLF(A,B,\sigma_W^2)=\ln f_X(\mathbf{x};A,B,\sigma_W^2)=-\frac{N}{2}\ln\left(2\pi\right)-\frac{N}{2}\ln\left(\sigma_W^2\right)-\frac{1}{2\sigma_W^2}\|\mathbf{x}-\mathbf{H\theta}\|^2$$

(2) *Derive the joint ML estimators of A and B when the noise power σ_W^2 is a priori known and provide a geometrical interpretation of them.*

$$\hat{\boldsymbol{\theta}}_{ML} = \left(\mathbf{H}^T\mathbf{H}\right)^{-1}\mathbf{H}^T\mathbf{X} = \begin{bmatrix} \|\mathbf{h}_1\|^2 & \mathbf{h}_1^T\mathbf{h}_2 \\ \mathbf{h}_1^T\mathbf{h}_2 & \|\mathbf{h}_2\|^2 \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} \mathbf{X} = \begin{bmatrix} N & 0 \\ 0 & \frac{N}{2} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} \mathbf{X} = \begin{bmatrix} \frac{1}{N}\mathbf{h}_1^T\mathbf{X} \\ \frac{2}{N}\mathbf{h}_2^T\mathbf{X} \end{bmatrix}$$

$$\hat{A}_{ML} = \frac{1}{N}\mathbf{h}_1^T\mathbf{X} = \frac{1}{N}\sum_{n=0}^{N-1} x[n],$$

$$\hat{B}_{ML} = \frac{2}{N}\mathbf{h}_2^T\mathbf{X} = \frac{2}{N}\sum_{n=0}^{N-1} x[n]\cos\left(2\pi F_0 n\right).$$

A and B are estimated at the output of two (orthogonal) linear matched filters: they are the components of the received signal $\mathbf{X}$ onto the 2D signal subspace, i.e. the 2D subspace generated by the two columns of $\mathbf{H}$. The two estimators are linear, so they are Gaussian distributed.

(3) *Derive bias and MSE of the ML estimators of A and B, establish if they are unbiased and consistent, derive which one of the two has better estimation accuracy and explain the reason. Then, plot the MSE of $\hat{B}_{ML}$ as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-3} when the noise power is $\sigma_W^2 = 1$.*

$$E\left\{\hat{A}_{ML}\right\} = \frac{1}{N}\mathbf{h}_1^T E\left\{\mathbf{X}\right\} = \frac{1}{N}\mathbf{h}_1^T\left(A\mathbf{h}_1 + B\mathbf{h}_2\right) = \frac{1}{N}\mathbf{h}_1^T\mathbf{h}_1 = A$$

$$E\left\{\hat{B}_{ML}\right\} = \frac{2}{N}\mathbf{h}_2^T E\left\{\mathbf{X}\right\} = \frac{2}{N}\mathbf{h}_2^T\left(A\mathbf{h}_1 + B\mathbf{h}_2\right) = \frac{2}{N}\mathbf{h}_2^T\mathbf{h}_2 = B$$

Both the estimators are unbiased.

$$MSE\left\{\hat{A}_{ML}\right\} = E\left\{\left(A-\hat{A}_{ML}\right)^2\right\} = E\left\{\left(A-\frac{1}{N}\mathbf{h}_1^T\mathbf{X}\right)^2\right\} = E\left\{\left(A-\frac{1}{N}\mathbf{h}_1^T\left(A\mathbf{h}_1+B\mathbf{h}_2+\mathbf{W}\right)\right)^2\right\}$$

$$= E\left\{\left(A-A-\frac{1}{N}\mathbf{h}_1^T\mathbf{W}\right)^2\right\} = E\left\{\left(-\frac{1}{N}\mathbf{h}_1^T\mathbf{W}\right)^2\right\} = \frac{1}{N^2}E\left\{\mathbf{h}_1^T\mathbf{W}\left(\mathbf{h}_1^T\mathbf{W}\right)^T\right\}$$

$$= \frac{1}{N^2}E\left\{\mathbf{h}_1^T\mathbf{W}\mathbf{W}^T\mathbf{h}_1\right\} = \frac{\sigma_W^2}{N^2}\mathbf{h}_1^T\mathbf{h}_1 = \frac{\sigma_W^2}{N}$$

$$MSE\left\{\hat{B}_{ML}\right\} = E\left\{\left(B-\hat{B}_{ML}\right)^2\right\} = E\left\{\left(-\frac{2}{N}\mathbf{h}_2^T\mathbf{W}\right)^2\right\} = \frac{4}{N^2}E\left\{\mathbf{h}_2^T\mathbf{W}\left(\mathbf{h}_2^T\mathbf{W}\right)^T\right\} = \frac{4}{N^2}E\left\{\mathbf{h}_2^T\mathbf{W}\mathbf{W}^T\mathbf{h}_2\right\}$$

$$= \frac{4\sigma_W^2}{N^2}\mathbf{h}_2^T\mathbf{h}_2 = \frac{2\sigma_W^2}{N}$$

Both the estimators are consistent.

$MSE\left\{\hat{A}_{ML}\right\} < MSE\left\{\hat{B}_{ML}\right\}$, the reason is that the energy of $\mathbf{h}_1$ is greater than the energy of $\mathbf{h}_2$, this allows to estimate *A* better than *B*.

$$\sigma_W^2 = 1: \quad MSE\left\{\hat{B}_{ML}\right\} = \frac{2}{N} \le 10^{-3} \quad \Rightarrow \quad N \ge 2000$$

(4) *Find the CRBs on the joint estimation of A and B and establish if the ML estimators previously derived are efficient or not.*

The derivative of the logarithm of the LR can be put in the following form:

$$\ln f_X(\mathbf{x}; A, B, \sigma_W^2) = -\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln\left(\sigma_W^2\right) - \frac{1}{2\sigma_W^2}\left((\mathbf{x}-\mathbf{H\theta})^T(\mathbf{x}-\mathbf{H\theta})\right)$$

$$\frac{\partial \ln L(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{1}{\sigma_W^2}\left[\mathbf{H}^T(\mathbf{x}-\mathbf{H\theta})\right] = \frac{1}{\sigma_W^2}\left[\mathbf{H}^T\mathbf{x}-\mathbf{H}^T\mathbf{H\theta}\right] = \frac{\left(\mathbf{H}^T\mathbf{H}\right)}{\sigma_W^2}\left[\left(\mathbf{H}^T\mathbf{H}\right)^{-1}\mathbf{H}^T\mathbf{x}-\boldsymbol{\theta}\right] = \mathbf{I}(\boldsymbol{\theta})\left[\mathbf{g}(\mathbf{x})-\boldsymbol{\theta}\right]$$

$$\begin{cases} FIM: \quad \mathbf{I}(\boldsymbol{\theta}) = \dfrac{\left(\mathbf{H}^T\mathbf{H}\right)}{\sigma_W^2} \quad \rightarrow \quad CRB(\boldsymbol{\theta}) = \sigma_W^2 \left(\mathbf{H}^T\mathbf{H}\right)^{-1} = \sigma_W^2 \begin{bmatrix} N & 0 \\ 0 & \dfrac{N}{2} \end{bmatrix}^{-1} = \begin{bmatrix} \dfrac{\sigma_W^2}{N} & 0 \\ 0 & \dfrac{2\sigma_W^2}{N} \end{bmatrix} \\ \hat{\boldsymbol{\theta}}_{ML} = \mathbf{g}(\mathbf{x}) = \left(\mathbf{H}^T\mathbf{H}\right)^{-1}\mathbf{H}^T\mathbf{x} = \mathbf{H}^{\#}\mathbf{x} \end{cases}$$

Hence, the estimator is efficient, in addition to be linear, Gaussian, unbiased and consistent.

$$MSE\left\{\hat{A}_{ML}\right\} = CRB(A) = \frac{\sigma_W^2}{N}, \quad MSE\left\{\hat{B}_{ML}\right\} = CRB(B) = \frac{2\sigma_W^2}{N}$$

Otherwise, the CRB can be derived following the usual procedure, from the inverse of the FIM. The FIM elements are obtained by calculating the 2nd order derivatives of the LLF, $LLF(A,B,\sigma_W^2) = \ln f_X(\mathbf{x};A,B,\sigma_W^2)$:

$$I_{11} = -E\left\{\frac{d^2 \ln f_X(\mathbf{x};A,B,\sigma_W^2)}{dA^2}\right\} = \frac{N}{\sigma_W^2}, \quad I_{22} = -E\left\{\frac{d^2 \ln f_X(\mathbf{x};A,B,\sigma_W^2)}{dB^2}\right\} = \frac{N}{2\sigma_W^2},$$

$$I_{12} = I_{21} = -E\left\{\frac{d^2 \ln f_X(\mathbf{x};A,B,\sigma_W^2)}{dAdB}\right\} = 0$$

Hence, $CRB(A) = \dfrac{\sigma_W^2}{N}, \quad CRB(B) = \dfrac{2\sigma_W^2}{N}$.

(5) *Assume now that also the noise power* σ_W^2 *is unknown: derive an unbiased estimator of* σ_W^2.

$$\frac{dLLF(\hat{A}_{ML},\hat{B}_{ML},\sigma_W^2)}{d\sigma_W^2} = \frac{d\ln f_X(\mathbf{x};\hat{A}_{ML},\hat{B}_{ML},\sigma_W^2)}{d\sigma_W^2} = -\frac{N}{2\sigma_W^2} + \frac{1}{2\sigma_W^4}\left\|\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}_{ML}\right\|^2 = 0$$

$$\hat{\sigma}_{W,ML}^2 = \frac{1}{N}\left\|\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}_{ML}\right\|^2 = \frac{1}{N}\left\|\mathbf{x} - \mathbf{H}\mathbf{H}^{\#}\mathbf{X}\right\|^2 = \frac{1}{N}\left\|(\mathbf{I} - \mathbf{H}\mathbf{H}^{\#})\mathbf{X}\right\|^2 = \frac{1}{N}\left\|(\mathbf{I} - \mathbf{P}_H)\mathbf{X}\right\|^2 = \frac{1}{N}\mathbf{X}^T(\mathbf{I} - \mathbf{P}_H)\mathbf{X}$$

The ML estimator projects the received data vector **X** onto the subspace orthogonal to the 2D signal subspace and then calculates the energy of the data after this projection, i.e. the sum of the powers of the *N*-2 components where only the noise is present. Since the signal subspace is 2-D, the orthogonal noise subspace is (N-2)-D, hence the mean $E\left\{\mathbf{X}^T(\mathbf{I}-\mathbf{P}_H)\mathbf{X}\right\}$ is equal to $(N-2)\sigma_W^2$.

Hence, if we want to have an unbiased estimator we should divide by N-2 instead of N:

$$\hat{\sigma}_{W,unbiased}^2 = \frac{1}{N-2}\mathbf{X}^T(\mathbf{I}-\mathbf{P}_H)\mathbf{X}$$